

THE COMPLETE WORKOUT & EXERCISE KNOWLEDGE BASE

Strength, Cardio, Flexibility, Splits & Goal-Based Programs
Gym, Home, and Equipment-Free Training · Beginner to Advanced

Reference document for AI fitness assistants and personal use.

Not a substitute for guidance from a certified trainer or doctor.

1. Types of Exercise — Overview

Fitness is generally built from four pillars: strength (resistance training), cardiovascular endurance, flexibility/mobility, and balance/stability. A well-rounded routine draws from all four, weighted toward whichever matches the individual's goal.

Category	Purpose	Examples
Strength / Resistance	Build muscle, increase strength, boost metabolism	Weightlifting, calisthenics, resistance bands
Cardiovascular	Heart/lung health, fat loss, endurance	Running, cycling, swimming, HIIT
Flexibility / Mobility	Joint range of motion, injury prevention	Static stretching, yoga, dynamic mobility drills
Balance / Stability	Core control, coordination, fall prevention	Single-leg work, stability ball, yoga balance poses

2. Warm-Up & Cool-Down Protocols

2.1 Warm-Up (5–10 minutes before training)

A proper warm-up raises body temperature, increases blood flow to muscles, and primes the nervous system — reducing injury risk and improving performance.

- **General warm-up (2–3 min):** light cardio — jogging in place, jumping jacks, cycling at low intensity.
- **Dynamic stretches (3–5 min):** leg swings, arm circles, walking lunges, hip openers, torso twists.
- **Movement-specific primer (2–3 min):** light sets of the first exercise you'll perform (e.g., empty-bar squats before heavy squats).

2.2 Cool-Down (5–10 minutes after training)

- Light cardio (2–3 min) to gradually lower heart rate.
- Static stretching (held 20–30 sec per muscle) for major muscles worked that day.
- Deep breathing to shift the body out of a 'training' state and into recovery.

Avoid long static stretching before heavy lifting — it can temporarily reduce strength output. Save deep static stretches for after training or separate sessions.

3. Strength Training — Exercise Library by Muscle Group

Compound exercises work multiple muscle groups and joints at once (most efficient, should form the base of most programs). Isolation exercises target a single muscle for extra volume or weak-point work.

3.1 Chest

Exercise	Type	Equipment
Barbell bench press	Compound	Barbell + bench
Dumbbell bench press	Compound	Dumbbells + bench
Incline bench press	Compound	Barbell/dumbbell + incline bench
Push-ups	Compound	Bodyweight
Chest dips	Compound	Dip bars
Cable/dumbbell fly	Isolation	Cables or dumbbells
Pec-deck machine	Isolation	Machine

3.2 Back

Exercise	Type	Equipment
Deadlift (conventional/sumo)	Compound	Barbell
Pull-ups / chin-ups	Compound	Pull-up bar
Bent-over barbell row	Compound	Barbell
Lat pulldown	Compound	Cable machine
Seated cable row	Compound	Cable machine
Single-arm dumbbell row	Compound	Dumbbell + bench
Straight-arm pulldown	Isolation	Cable machine
Face pulls	Isolation (rear delt/upper back)	Cable + rope attachment

3.3 Shoulders

Exercise	Type	Equipment
Overhead press (barbell/dumbbell)	Compound	Barbell/dumbbell
Arnold press	Compound	Dumbbells
Lateral raise	Isolation	Dumbbells/cables
Front raise	Isolation	Dumbbells/plate
Rear-delt fly	Isolation	Dumbbells/cables
Shrugs	Isolation (traps)	Barbell/dumbbells

3.4 Arms — Biceps & Triceps

Exercise	Target	Equipment
Barbell/EZ-bar curl	Biceps	Barbell
Dumbbell curl (standing/seated)	Biceps	Dumbbells
Hammer curl	Biceps (brachialis)	Dumbbells
Preacher curl	Biceps	Preacher bench + bar
Close-grip bench press	Triceps (compound)	Barbell
Tricep pushdown (cable)	Triceps	Cable machine
Overhead tricep extension	Triceps	Dumbbell/cable
Dips (triceps-focused)	Triceps	Dip bars

3.5 Legs

Exercise	Type	Equipment
Barbell back squat	Compound	Barbell + rack
Front squat	Compound	Barbell + rack
Romanian deadlift	Compound (hamstrings/glutes)	Barbell/dumbbells
Leg press	Compound	Machine
Walking lunges	Compound	Bodyweight/dumbbells
Bulgarian split squat	Compound	Dumbbells + bench
Leg extension	Isolation (quads)	Machine
Leg curl (lying/seated)	Isolation (hamstrings)	Machine
Calf raise (standing/seated)	Isolation (calves)	Machine/bodyweight
Hip thrust	Compound (glutes)	Barbell + bench

3.6 Core

Exercise	Focus	Equipment
Plank (standard/side)	Isometric core stability	Bodyweight
Hanging leg raise	Lower abs	Pull-up bar
Cable woodchopper	Rotational core/obliques	Cable machine
Russian twist	Obliques	Bodyweight/weight plate
Ab wheel rollout	Full core, advanced	Ab wheel
Dead bug	Core control, beginner-friendly	Bodyweight
Hollow body hold	Core stability	Bodyweight

4. Bodyweight & Calisthenics Training

Calisthenics uses body weight as resistance — ideal for home workouts, travel, or building relative strength and control. Progression is achieved by changing leverage, range of motion, or adding reps/tempo rather than external load.

Movement Pattern	Beginner	Intermediate	Advanced
Push	Knee push-ups	Standard push-ups	Archer / one-arm push-ups
Pull	Negative pull-ups / inverted rows	Pull-ups / chin-ups	Weighted or one-arm pull-ups
Squat	Bodyweight squats	Jump squats / Bulgarian split squat	Pistol squats
Hinge	Glute bridges	Single-leg Romanian deadlift (bodyweight)	Nordic curls
Core	Dead bug / plank	Hanging knee raise	Front lever progressions
Push (vertical)	Pike push-ups	Elevated pike push-ups	Handstand push-ups

5. Cardiovascular Training

5.1 Types of Cardio

Type	Intensity	Example	Best For
LISS (Low-Intensity Steady State)	Low, sustained	Brisk walking, easy cycling	Recovery, fat oxidation, beginners
MISS (Moderate-Intensity Steady State)	Moderate, sustained	Jogging, swimming laps	General cardio fitness
HIIT (High-Intensity Interval Training)	Alternating high/low	Sprints, burpee intervals, cycling sprints	Time-efficient conditioning, VO2 max
Steady-state endurance	Moderate, long duration	Long-distance running, cycling	Endurance events, stamina building
Circuit training	Moderate-high, varied	Station-based bodyweight/weight circuits	Combined strength + cardio

5.2 Sample HIIT Protocol (Beginner-Friendly)

- Warm up 5 minutes (easy pace).
- 30 seconds high effort (e.g., fast cycling, sprint, jump squats) + 60–90 seconds easy recovery.
- Repeat for 6–8 rounds.
- Cool down 5 minutes.

As fitness improves, the work:rest ratio can shift toward more work and less rest (e.g., 40:20 or 1:1).

6. Flexibility & Mobility

6.1 Static vs Dynamic Stretching

Type	What It Is	When to Use
Static stretching	Hold a stretch position 15–60 seconds, no movement	After workouts, or standalone flexibility sessions
Dynamic stretching	Controlled movement through a range of motion	Before workouts, as part of warm-up
PNF stretching	Contract-relax technique with a partner or band	Advanced flexibility work, often with guidance
Foam rolling (self-myofascial release)	Rolling pressure on muscles to release tightness	Before or after training, or on rest days

6.2 Yoga as Flexibility & Mobility Training

Yoga combines flexibility, controlled breathing, and light strength/balance work. Common styles range from gentle (Hatha, Yin) to vigorous (Vinyasa, Ashtanga, Power Yoga). Even 10–15 minutes a few times a week can meaningfully improve mobility and reduce stiffness, and it pairs well alongside strength training.

7. Workout Splits — How to Structure a Week

Split	Typical Frequency	Best For
Full Body	3x/week (e.g., Mon/Wed/Fri)	Beginners, limited time, general fitness
Upper/Lower	4x/week	Intermediate lifters wanting balanced frequency
Push/Pull/Legs (PPL)	3x or 6x/week	Intermediate–advanced, muscle-focused training
Bro Split (one muscle/day)	5x/week	Advanced lifters prioritizing per-muscle volume
Functional/Circuit	3–5x/week	General fitness, fat loss, varied conditioning

7.1 Example — Push/Pull/Legs (6-Day) Layout

Day	Focus
Day 1	Push (chest, shoulders, triceps)
Day 2	Pull (back, biceps)
Day 3	Legs (quads, hamstrings, glutes, calves)
Day 4	Push (variation)
Day 5	Pull (variation)
Day 6	Legs (variation)
Day 7	Rest / active recovery

7.2 Example — Full Body (3-Day) Layout for Beginners

Day	Example Exercises
Day 1	Squat, bench press, bent-over row, plank
Day 2	Rest or light cardio/mobility
Day 3	Deadlift, overhead press, lat pulldown, hanging leg raise
Day 4	Rest or light cardio/mobility
Day 5	Leg press, incline press, seated cable row, dead bug
Day 6–7	Rest / light activity

8. Training Variables — Sets, Reps, Rest & Progression

Goal	Reps per Set	Sets	Rest Between Sets
Strength (max force)	1–6	3–5	2–5 minutes
Hypertrophy (muscle size)	6–12	3–4	60–90 seconds
Muscular endurance	12–20+	2–3	30–60 seconds
Power (explosiveness)	1–5 (explosive)	3–5	2–5 minutes

8.1 Progressive Overload

Muscles adapt to demand placed on them — progressive overload means gradually increasing that demand over time so the body keeps adapting. This can be done by:

- Adding weight to the bar/dumbbells.
- Adding reps or sets at the same weight.
- Improving range of motion or form quality.
- Reducing rest time between sets (for endurance/conditioning goals).
- Increasing training frequency for a muscle group.

8.2 RPE (Rate of Perceived Exertion)

A simple 1–10 scale for how hard a set felt. RPE 10 means no reps left in the tank; RPE 7–8 means 2–3 reps left. Many well-designed programs target RPE 7–9 on most working sets, reserving RPE 10 for occasional testing, since training to failure every set increases fatigue and injury risk without proportionally better results.

9. Sample Weekly Programs by Goal

General templates — adjust volume/intensity to individual experience level and recovery capacity.

9.1 Fat Loss (4 Days/Week, Full Body + Cardio)

Day	Session
Mon	Full body strength (squat, row, push-up, plank) + 15 min incline walk
Tue	20–25 min HIIT (bike/bodyweight circuit)
Wed	Full body strength (deadlift, overhead press, lunges, core)
Thu	Rest or light walk/yoga
Fri	Full body strength + 20 min steady cardio
Sat	30–40 min LISS cardio (walk/cycle/swim)
Sun	Rest

9.2 Muscle Gain (5 Days/Week, Upper/Lower + PPL hybrid)

Day	Session
Mon	Upper body (bench, row, overhead press, curls/pushdowns)
Tue	Lower body (squat, RDL, leg press, calves)
Wed	Rest or light mobility
Thu	Push (incline press, lateral raise, triceps)
Fri	Pull (deadlift, pull-ups, face pulls, biceps)
Sat	Legs + core (lunges, leg curl, hanging raise, plank)
Sun	Rest

9.3 General Fitness / Beginner (3 Days/Week)

Day	Session
Mon	Full body strength (bodyweight or light weights) + 10 min mobility
Tue	Rest or 20–30 min walk
Wed	Full body strength + light cardio finisher
Thu	Rest or yoga/stretching
Fri	Full body strength + 15–20 min cardio of choice
Sat–Sun	Active recovery — walking, light activity, stretching

10. Equipment Guide — Home vs Gym

Setting	Core Equipment	Good For
No equipment	Just bodyweight	Push-ups, squats, lunges, planks, burpees — great starting point
Minimal home gym	Resistance bands, one pair of adjustable dumbbells, pull-up bar	Covers most muscle groups with progressive options
Full home gym	Barbell + plates, rack, bench, dumbbells	Near-complete programming flexibility
Commercial gym	Full range of machines, free weights, cardio equipment	Maximum exercise variety and load options

11. Common Mistakes & Injury Prevention

- **Skipping warm-ups** — increases injury risk, especially for heavy compound lifts.
- **Ego lifting (too much weight, poor form)** — the most common cause of avoidable injury; prioritize technique over load.
- **Doing too much too soon** — sudden large jumps in volume/intensity are a common cause of overuse injuries.
- **Neglecting rest days** — muscles grow and repair during rest, not just during training.
- **Ignoring pain signals** — 'muscle fatigue' is normal; sharp or joint pain is a sign to stop and reassess.
- **Poor sleep** — undermines recovery, performance, and injury resilience regardless of how well training is programmed.
- **Inconsistent programming** — constantly switching routines before giving one enough time (4–8+ weeks) to show results.

12. Recovery — Sleep, Rest Days & DOMS

12.1 Why Recovery Matters

Training creates the stimulus for adaptation; recovery is when the actual adaptation (muscle repair, strength gains, energy replenishment) happens. Without adequate recovery, progress stalls and injury risk rises, regardless of how well-designed the training program is.

12.2 Sleep

- 7–9 hours per night is the general target for most adults.
- Growth hormone release and muscle repair are concentrated in deep sleep stages.
- Poor sleep is linked to reduced strength performance, slower recovery, and higher injury risk.

12.3 DOMS (Delayed Onset Muscle Soreness)

Soreness appearing 24–72 hours after unfamiliar or intense exercise. It is normal, especially when starting a new program or increasing intensity, and is not a requirement for progress — training can be effective without significant soreness every session.

- Light movement, walking, and gentle stretching can ease DOMS.
- Staying hydrated and maintaining adequate protein intake supports faster recovery.
- Sharp or one-sided joint pain is different from DOMS and should not be pushed through.

12.4 Rest Days

At least 1–2 full or active-recovery rest days per week is typical for most training programs. Active recovery (walking, light yoga, easy cycling) can be more effective than complete inactivity for reducing stiffness and maintaining the exercise habit.

13. Quick-Reference Cheat Sheet

Topic	Quick Takeaway
Warm-up	5–10 min: light cardio + dynamic stretching + movement-specific primer
Strength rep range	1–6 reps for strength, 6–12 for muscle size, 12–20+ for endurance
Rest between sets	2–5 min for strength/power, 60–90 sec for hypertrophy
Progressive overload	Add weight, reps, sets, or reduce rest over time — don't stay static
Cardio types	LISS for recovery/fat oxidation, HIIT for time-efficient conditioning
Beginner split	Full body, 3x/week is the simplest effective starting point
Advanced split	PPL (6-day) or Upper/Lower (4-day) for more per-muscle frequency
Flexibility timing	Dynamic stretching before training, static stretching after
Sleep	7–9 hours/night — critical for recovery and performance
Rest days	At least 1–2 per week, active recovery is often better than total rest
Consistency	A program followed consistently for 8+ weeks beats constant switching

This document is a general fitness reference for informational purposes. It is not personalized medical or training advice. Anyone with an existing injury, heart condition, pregnancy, or other health concern should consult a doctor or certified trainer before starting a new exercise program.